CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

# Chapter 13: Section and Mid-Point Formula

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

## 1 Centroid of a triangle

**1.** Triangle $ABC$ has vertices $A(0, 6)$, $B(4, -4)$ and $C(-4, -2)$. $D$ is the point where $AC$ crosses the $x$-axis. (a) Find the coordinates of $D$. (b) Find the centroid of $\triangle ABC$. (c) $D$ divides $AC$ in the ratio $k : 1$; find $k$. (d) Find the equation of the line $BD$. ICSE 2024, 4 marks

SOLUTION

(c) Let $D$ divide $AC$ in the ratio $k : 1$. Its $y$-coordinate is 0: $\dfrac{-2k + 6}{k + 1} = 0$, so $k = 3$.

(a) Then $x = \dfrac{3(-4) + 0}{3 + 1} = -3$, so $D = (-3, 0)$.

(b) Centroid $= \left(\dfrac{0 + 4 - 4}{3}, \dfrac{6 - 4 - 2}{3}\right) = (0, 0)$

(d) Slope of $BD = \dfrac{0 - (-4)}{-3 - 4} = -\dfrac{4}{7}$. Through $D$: $y = -\dfrac{4}{7}(x + 3)$, so $7y = -4x - 12$.

**Answer: (a)** $D(-3, 0)$ **(b)** $(0, 0)$ **(c)** $k = 3$ **(d)** $4x + 7y + 12 = 0$

**2.** The vertices of $\triangle ABC$ are $(-4, -2)$, $(6, 2)$ and $(4, 6)$. Find its centroid. (a) $(2, 2)$ (b) $(2, 3)$ (c) $(3, 3)$ (d) $(0, -1)$ ICSE 2023, 1 mark

SOLUTION

Centroid $= \left(\dfrac{-4 + 6 + 4}{3}, \dfrac{-2 + 2 + 6}{3}\right) = (2, 2)$

**Answer: (a)**

## 2 The section formula

**3.** $A(4, 0)$ lies on the $x$-axis and $B(0, 4)$ lies on the $y$-axis. $P$ is the point on $AB$ with $AP : PB = 3 : 1$. (a) Use the section formula to find $P$. (b) Find the equation of the line through $P$ perpendicular to $AB$. ICSE 2023, 4 marks

SOLUTION

(a) $P = \left(\dfrac{3 \times 0 + 1 \times 4}{3 + 1}, \dfrac{3 \times 4 + 1 \times 0}{3 + 1}\right) = (1, 3)$

(b) Slope of $AB = \dfrac{4 - 0}{0 - 4} = -1$, so the perpendicular line has slope 1.

$y - 3 = 1(x - 1)$

**Answer: (a)** $P(1, 3)$ **(b)** $y = x + 2$

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**4.** $M$ lies on the $x$-axis and $N$ lies on the $y$-axis. The point $P(3, 2)$ divides $MN$ in the ratio $2 : 3$. Find (a) the coordinates of $M$ and $N$ (b) the slope of $MN$. ICSE 2019, 3 marks

SOLUTION

(a) Let $M = (a, 0)$ and $N = (0, b)$. Then $P = \left(\frac{2 \times 0 + 3a}{5}, \frac{2b + 3 \times 0}{5}\right)$.

$\frac{3a}{5} = 3$ gives $a = 5$, and $\frac{2b}{5} = 2$ gives $b = 5$.

(b) Slope of $MN = \frac{5 - 0}{0 - 5} = -1$

**Answer: (a)** $M(5, 0)$, $N(0, 5)$ **(b) slope** $= -1$

**5.** $P(1, -2)$ lies on the segment joining $A(3, -6)$ and $B(x, y)$, with $AP : PB = 2 : 3$. Find the coordinates of $B$. ICSE 2017, 4 marks

SOLUTION

$1 = \frac{2x + 3(3)}{5}$, so $2x = -4$ and $x = -2$.

$-2 = \frac{2y + 3(-6)}{5}$, so $2y = 8$ and $y = 4$.

**Answer:** $B(-2, 4)$

## 3 Finding the ratio

**6.** In what ratio does the $y$-axis divide the segment joining $P(5, 3)$ and $Q(-5, 3)$? Find the point of intersection. ICSE 2020, 4 marks

SOLUTION

Let the ratio be $k : 1$. On the $y$-axis the $x$-coordinate is 0: $\frac{-5k + 5}{k + 1} = 0$, so $k = 1$.

The point is the mid-point: $\left(0, \frac{3 + 3}{2}\right) = (0, 3)$.

**Answer: ratio** $1 : 1$**; the point is** $(0, 3)$

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